

Quick Activity 1: School Supplies Calculator

A notebook costs ₱45, and a pen costs ₱12. A student buys several notebooks and pens and pays a certain amount. Develop an algorithm that calculates the total cost and change.

Assume that quantities are positive whole numbers and the payment is enough to cover the purchase.

Decomposition

Input	Process	Output
Get the quantity of notebooks and pens, cash amount paid	Calculate the notebook subtotal, pen subtotal, combined total, change	Show the total cost and change

Pseudocode

START

```
READ notebooks, pens, payment
notebook_price ← notebooks × 45
pen_price ← pens × 12
total_bill ← notebook_price + pen_price
change_due ← payment - total_bill
DISPLAY total_bill
DISPLAY change_due
```

END

Dry run calculations

Test Case 1

Input:

Notebooks = 3

Pens = 2

Payment = ₱200

Step	Variable	Calculation	Solution
1	Notebook Subtotal	3×45	₱135
2	Pen subtotal	2×12	₱24
3	Final bill	$135 + 24$	₱159
4	Change returned	$200 - 159$	₱41

Output: Total cost = ₱159, Change = ₱41

Test Case 2

Input:

Notebooks = 2

Pens = 3

Payment = ₱150

Step	Variable	Calculation	Solution
1	Notebook Subtotal	2×45	₱90
2	Pen subtotal	3×12	₱36
3	Final bill	$90 + 36$	₱126
4	Change returned	$150 - 126$	₱24

Output:

Total cost = ₱126, Change = ₱24